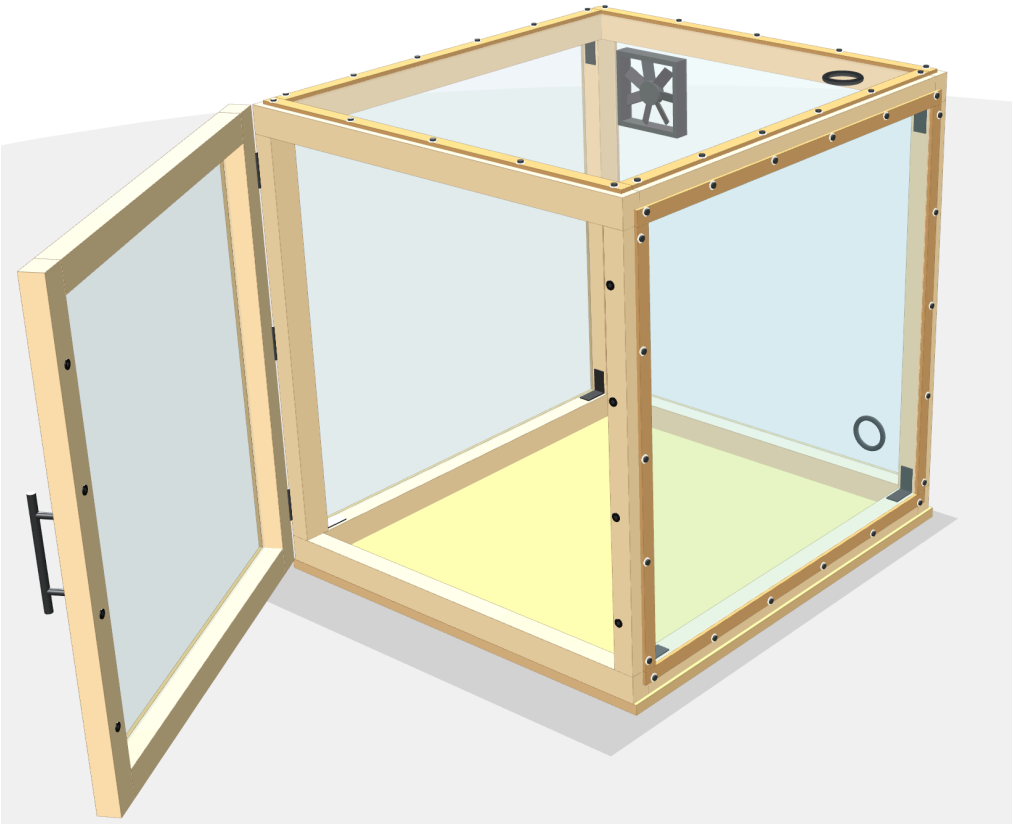
Bambu Lab A1 Wooden Enclosure

Timber machining drawings • external feed (AMS lite outside) + rebate-and-bead version



3D model view (door half open, printer hidden)

Frame (outside of wood)	**640 × 700 × 680**
Clear inside W×D×H	**580 × 640 × 640**
Finished size (door, beads, feet)	**about 650 × 743 × 710**
30×40 strip total	**10.0 m (6 × 2 m)**
20×5 bead total	**12.4 m (7 × 2 m)**

General notes

1. All dimensions in millimetres (mm).
2. Timber: 30×40 kiln-dried spruce or pine, planed four sides. Cut the rebates yourself with a router or use a joinery shop that can do stopped rebates; most timber yards only cut to length, in which case use the plain-strip drawings.
3. Tolerances: length ±1, section ±0.5; rebate width 10 (+0.5 / 0), depth 4 (±0.3).
4. Every rebate is 10 wide × 4 deep, so a 3 mm panel on 1 mm foam sits flush with the wood.
5. Profiles (sheet 3): P0 no rebate; PA one rebate in a 40 face; PB one rebate in a 30 face; PC two rebates, one in a 40 face and one in a 30 face, on diagonally opposite corners.
6. "Full length" = the rebate runs end to end. "Stopped" = solid wood is left at both ends; square the ends with a chisel.
7. Cut to length first, then rebate. Pencil the part number (A1, B2 …) on every piece.
8. Orange hatching = rebate; blue dashed outline = clear panel.

Sheet list

1. Overview
2. Cut list
3. Profiles and panel detail
4. Frame parts
5. Door frame
6. Frame assembly views
7. Clear panels
8. Appendix: screw-on version

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

Overview

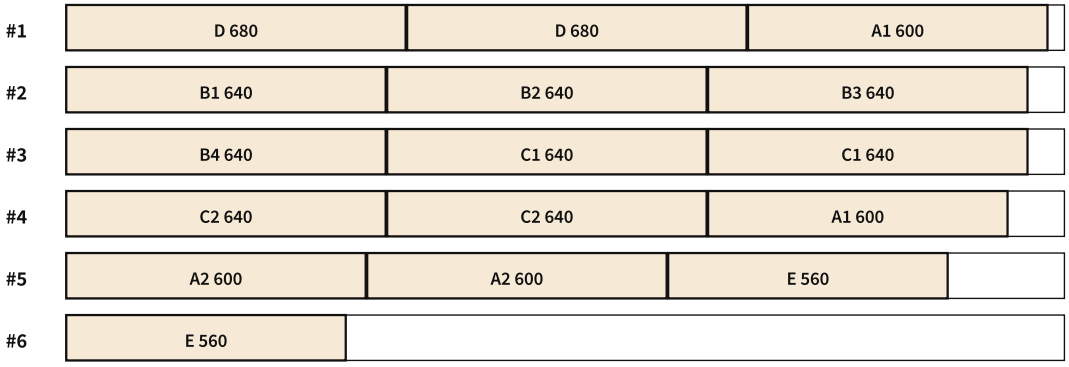
Sheet	Scale	Units	Date	Rev.
1 / 8	—	mm	2026-11-08	v2

Cut list • 30×40 strip

ID	Part	Size (length)	Profile	Rebate	Qty	Location
A1	Front post	30×40×600	PA	Full length	2	Front left & right
A2	Rear post	30×40×600	PC	Full length (×2)	2	Rear left & right
B1	Bottom front rail	30×40×640	P0	—	1	Bottom frame, front
B2	Bottom back rail	30×40×640	PA	Stopped 20–620	1	Bottom frame, back
B3	Top front rail	30×40×640	PB	Stopped 20–620	1	Top frame, front
B4	Top back rail	30×40×640	PC	Stopped 20–620 (×2)	1	Top frame, back
C1	Bottom side rail	30×40×640	PA	Full length	2	Bottom frame, left & right
C2	Top side rail	30×40×640	PC	Full length (×2)	2	Top frame, left & right
D	Door stile	30×40×680	PA	Stopped 30–650	2	Door left & right
E	Door rail	30×40×560	PA	Full length	2	Door top & bottom
F	Base board	650×705×15	—	—	1	Paulownia panel, cut to size only

Strip total 10 000 mm, 16 pieces (A1–E), plus 1 base board.

Cutting plan • 2 m stock, 3 mm kerf

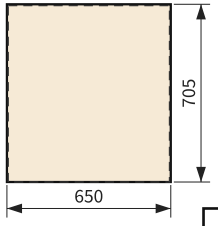


G Glazing bead • solid wood 20×5

Length	Qty	Use
660	4	Side panels, top & bottom (2 panels)
640	2	Top panel, left & right
620	4	Back panel top/bottom 2; top panel front/back 2
600	10	Side panels front/back 4; back panel sides 2; door all 4

Total 12.4 m, buy 7 × 2 m. Paulownia works; ready-made pine or beech strip splits less.
Horizontal beads run across the corners; upright beads fit between them.

F Base board 1:30



15 mm paulownia panel. Dashed = frame footprint (5 in at left, right and back; flush at the front).

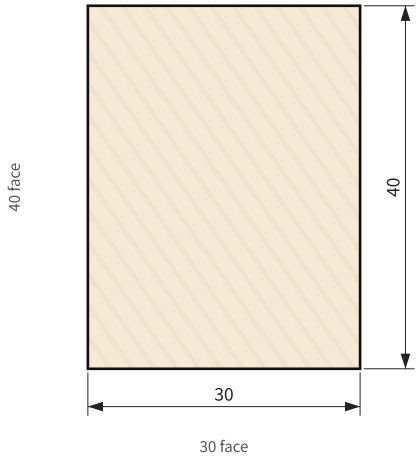
Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

Cut list

Sheet	Scale	Units	Date	Rev.
2 / 8	—	mm	2026-11-08	v2

Profiles (scale 1.2:1)

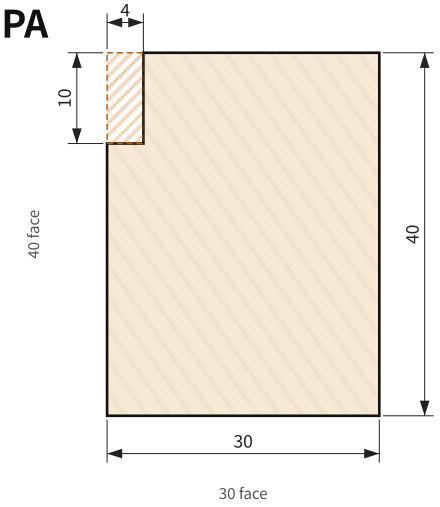
P0



No rebate

Used for: B1

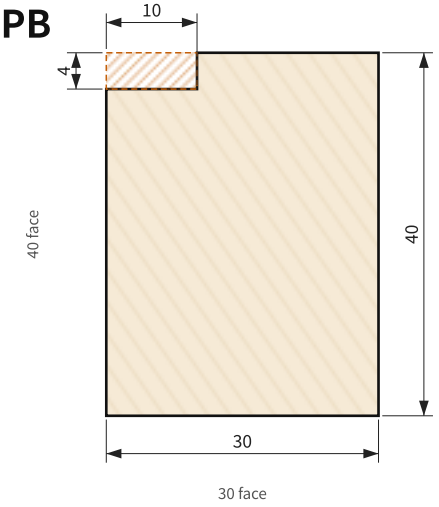
PA



One rebate in a 40 face: 10 wide, 4 deep

Used for: A1 B2 C1 D E

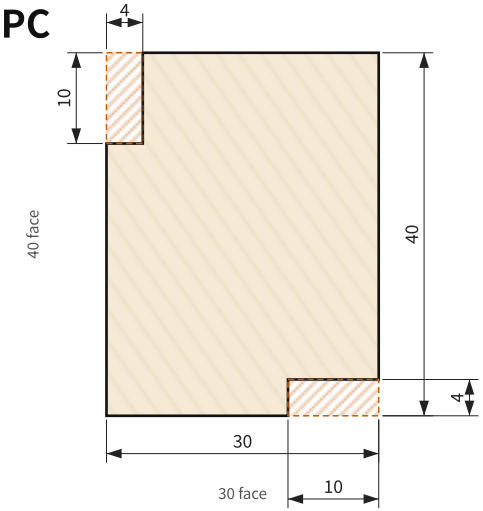
PB



One rebate in a 30 face: 10 wide, 4 deep

Used for: B3

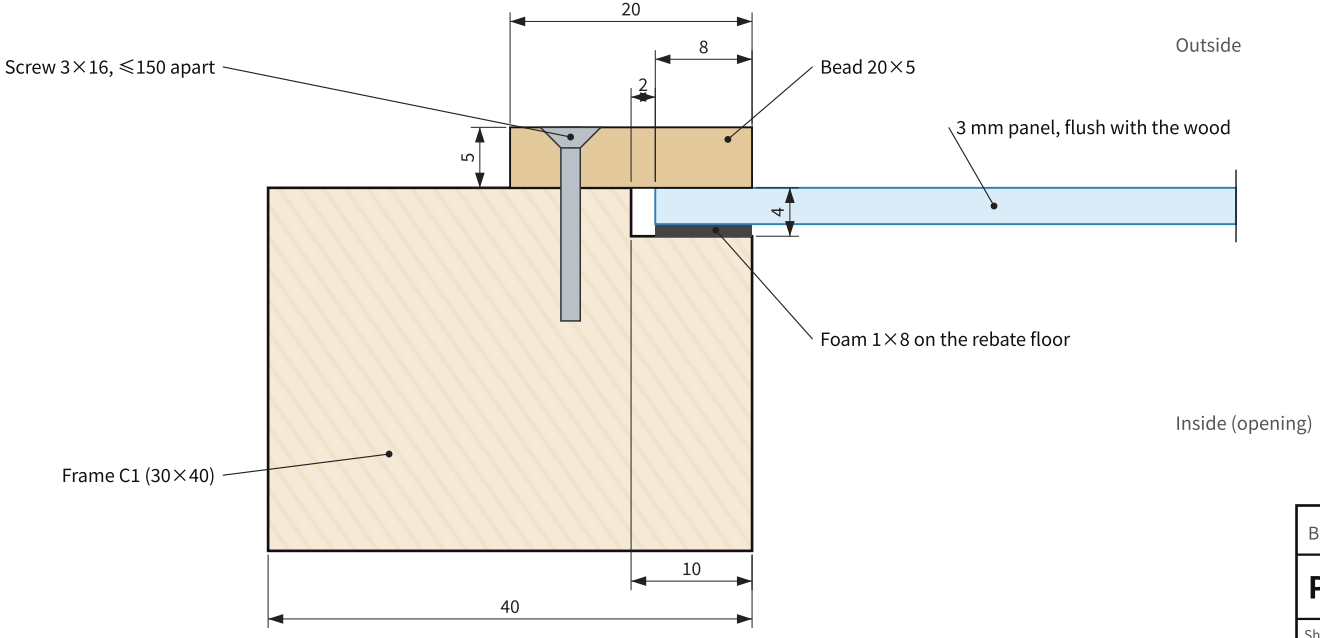
PC



Two: one in a 40 face + one in a 30 face, diagonal corners

Used for: A2 B4 C2

Panel detail (scale 1.6:1, bottom side rail C1 shown; every rebate works the same way)



2 mm = expansion gap: the panel grows when warm, so it must not be wedged in.

8 mm = how far the panel edge sits in the rebate.

Centre the panel with 2 mm spacers, then remove them once the beads are on.

Bambu Lab A1 enclosure · external feed · rebate + bead (router needed)

Profiles and panel detail

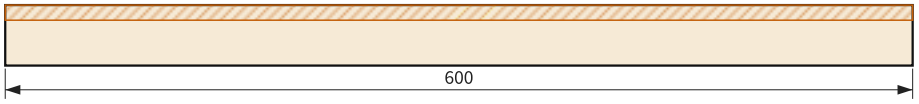
Sheet	Scale	Units	Date	Rev.
3 / 8	1.2:1 / 1.6:1	mm	2026-11-08	v2

Frame parts (A1–C2)

Scale 1:5. Bar = the rebated face; dashed = the second rebate on the opposite corner (far side); small drawing = section.

A1 Front post

×2 • PA

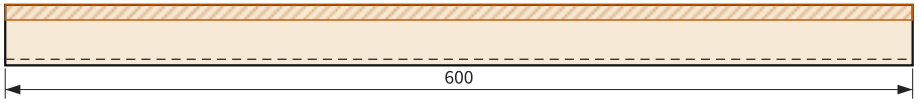


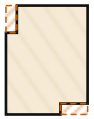


40 face outward (side), rebate on the rear edge. Left and right are the same part.

A2 Rear post

×2 • PC

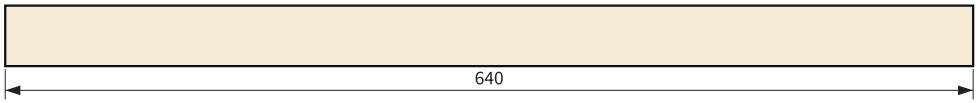




40 face outward (side), rebate on the front edge; 30 face to the back, rebate on the inner edge. Left/right identical, turned end for end.

B1 Bottom front rail

×1 • P0

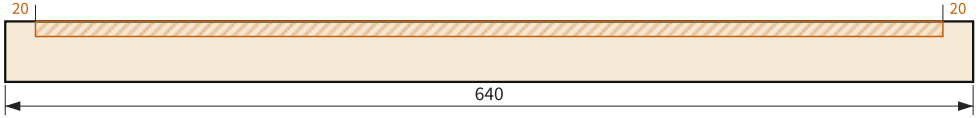




No rebate.

B2 Bottom back rail

×1 • PA

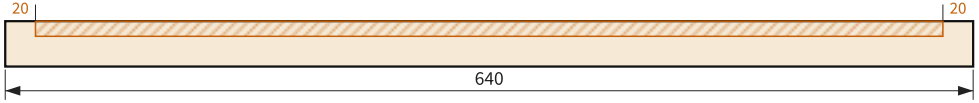




40 face to the back, rebate on the top edge; stopped, 20 solid at each end.

B3 Top front rail

×1 • PB

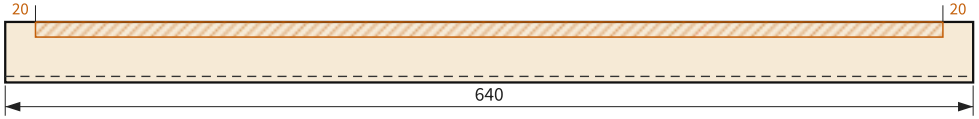


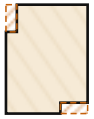


30 face up, rebate on the inner (rear) edge; stopped, 20 solid at each end.

B4 Top back rail

×1 • PC

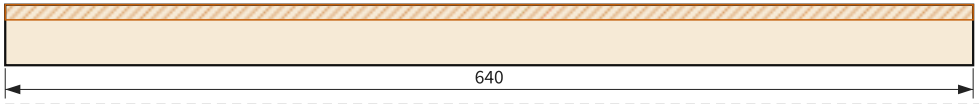




40 face to the back, rebate on the bottom edge; 30 face up, rebate on the inner (front) edge. Both stopped, 20 solid at each end.

C1 Bottom side rail

×2 • PA

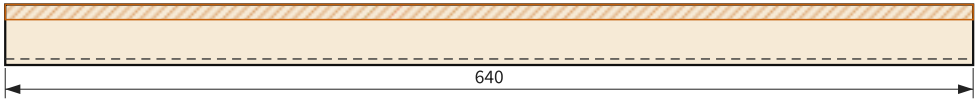


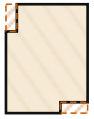


40 face outward, rebate on the top edge; full length.

C2 Top side rail

×2 • PC





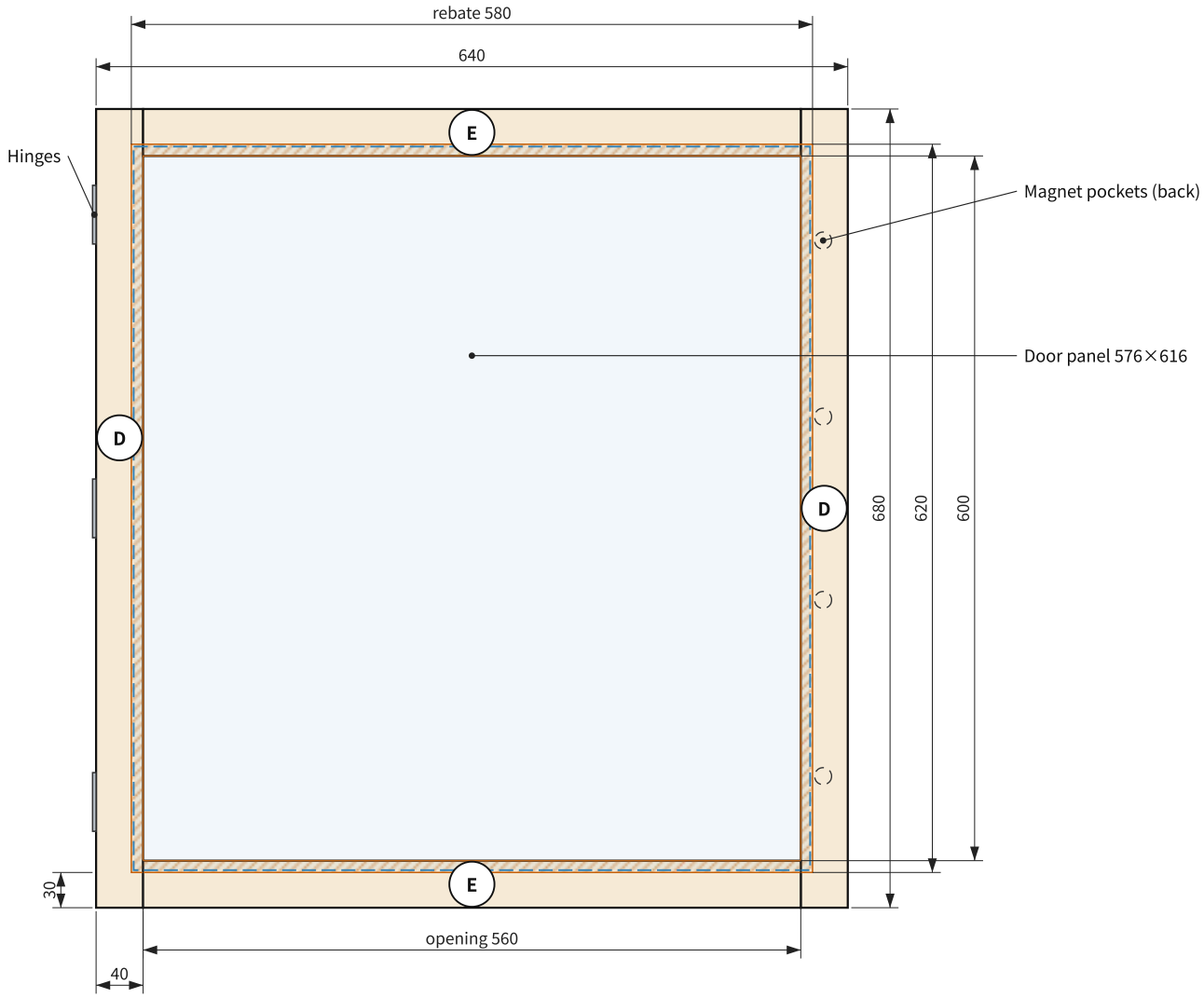
40 face outward, rebate on the bottom edge; 30 face up, rebate on the inner edge; full length.

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

Frame parts

Sheet	Scale	Units	Date	Rev.
4 / 8	1:5	mm	2026-11-08	v2

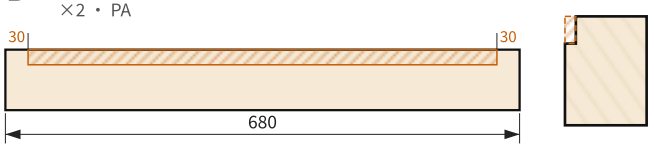
Door frame, front view (from outside, scale 1:6)



The rebate is one continuous ring, 580×620 outside / 560×600 inside. Stiles D are stopped (30–650); rails E run full length; they meet at the corners. Magnet pockets and hinges are DIY, not shop work.

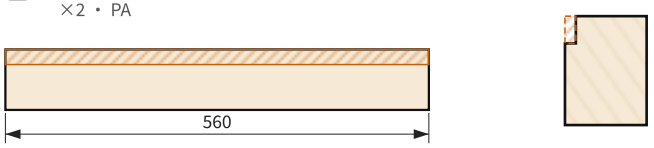
Parts

D Door stile
×2 • PA



40 face to the front, rebate on the inner edge; stopped 30–650, 30 solid at each end.

E Door rail
×2 • PA



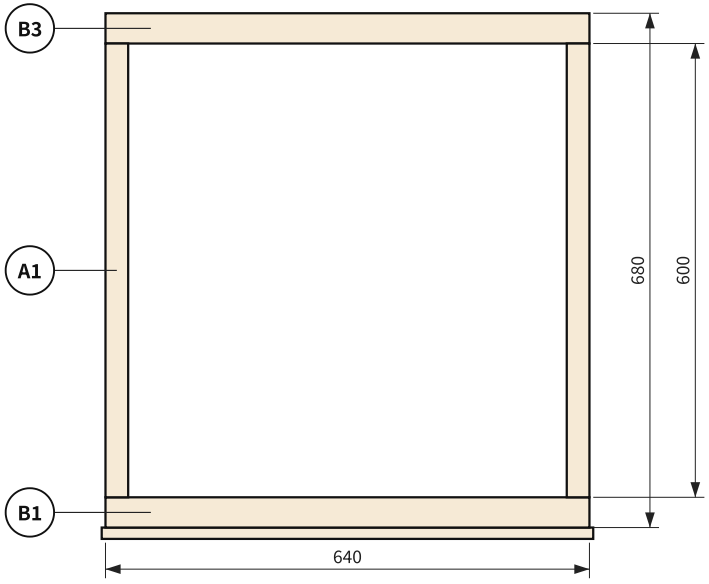
40 face to the front, rebate on the inner edge; full length.

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

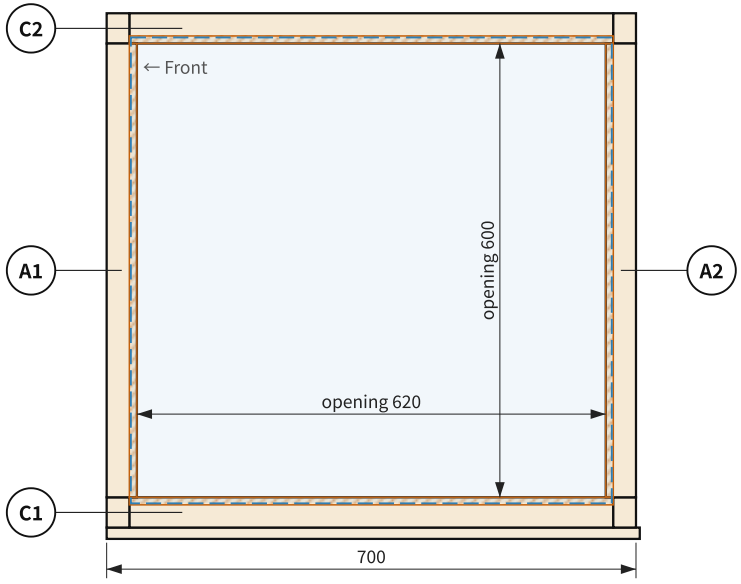
Door frame

Sheet	Scale	Units	Date	Rev.
5 / 8	1:6	mm	2026-11-08	v2

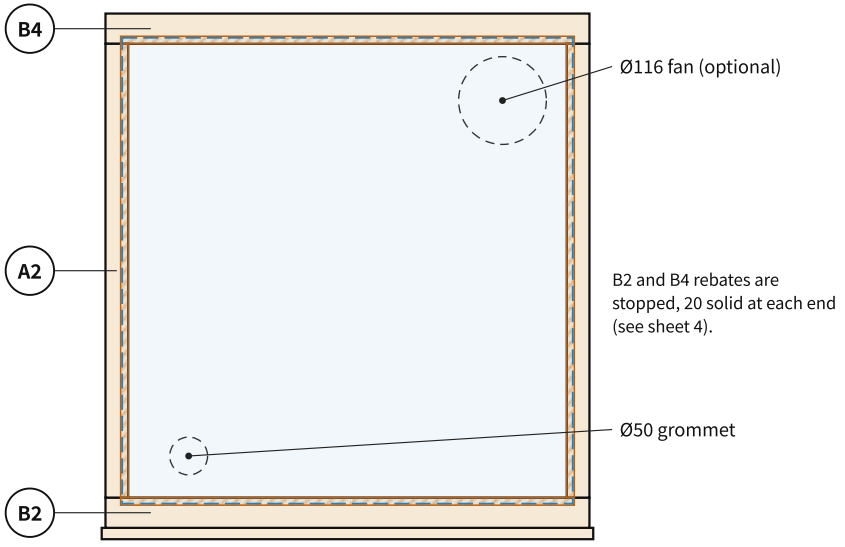
Front view (door removed)



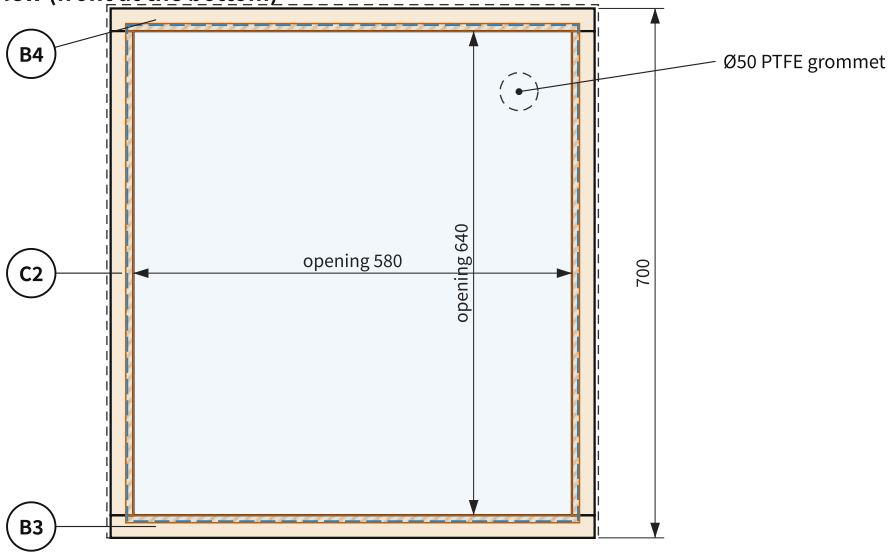
Right side view (front on the left; left side mirrors it)



Back view (from outside)



Top view (front at the bottom)



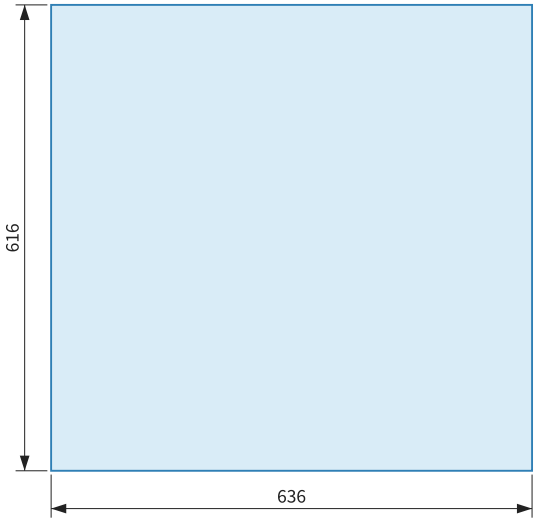
Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

Frame assembly views

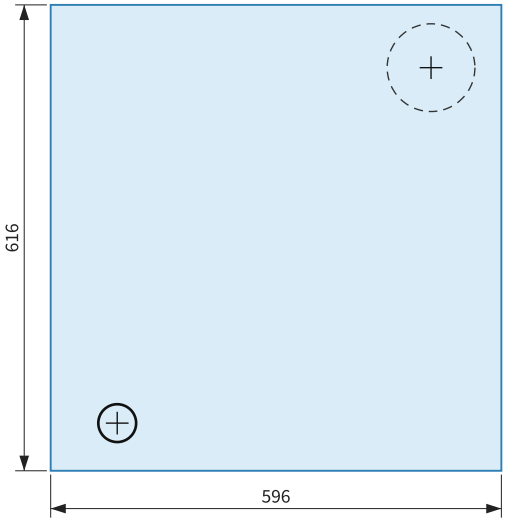
Sheet	Scale	Units	Date	Rev.
6 / 8	1:10	mm	2026-11-08	v2

Clear panels (rebate version, scale 1:10, seen from outside)

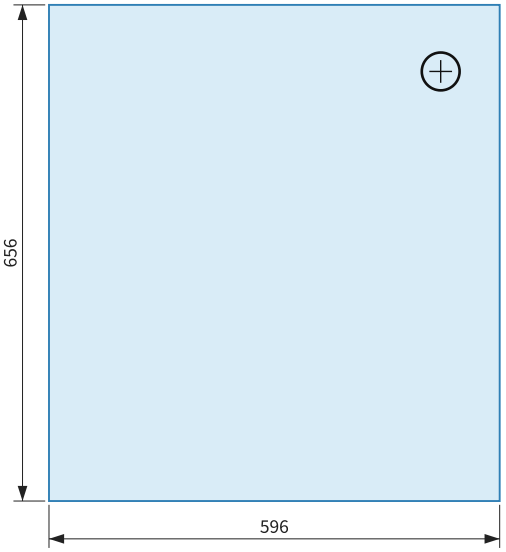
P1 Side panel (L/R) ×2



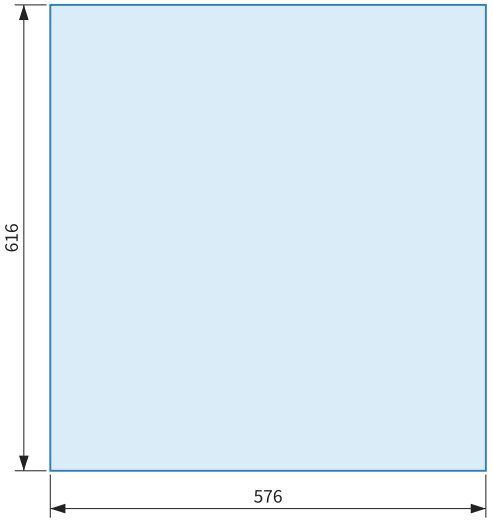
P2 Back panel ×1



P3 Top panel ×1



P4 Door panel ×1



Panel	Hole	Position (to hole centre)
P2	Ø50 grommet hole	88 from the left edge, 63 from the bottom (seen from outside; power-socket side)
P2	Ø116 fan hole (optional)	93 from the right edge, 83 from the top
P3	Ø50 grommet hole	78 from the right edge, 88 from the top (rear) edge; top of the drawing = back of the box

- Material: 3 mm cast acrylic or solid polycarbonate (solid PC recommended for the door). Not twin-wall PC: it is not see-through.
- Size tolerance ±0.5, polished edges, protective film left on. No screw holes around the edges.
- Hole positions can be adjusted to suit the A1 power socket and the AMS lite. Drill the door-handle holes once you have the handle.

Bambu Lab A1 enclosure · external feed · rebate + bead (router needed)

Clear panels

Sheet	Scale	Units	Date	Rev.
7 / 8	1:10	mm	2026-11-08	v2

Appendix: screw-on version (no rebates)

Use this version if you have no router and the timber shop cannot cut rebates: no strip is rebated, and the panels lie on the outside of the frame, fixed with ST4×25 screws and large flat washers (for 4 mm twin-wall sheet see the plain-strip drawings). Only this sheet differs; the other sheets still apply (ignore the rebates).

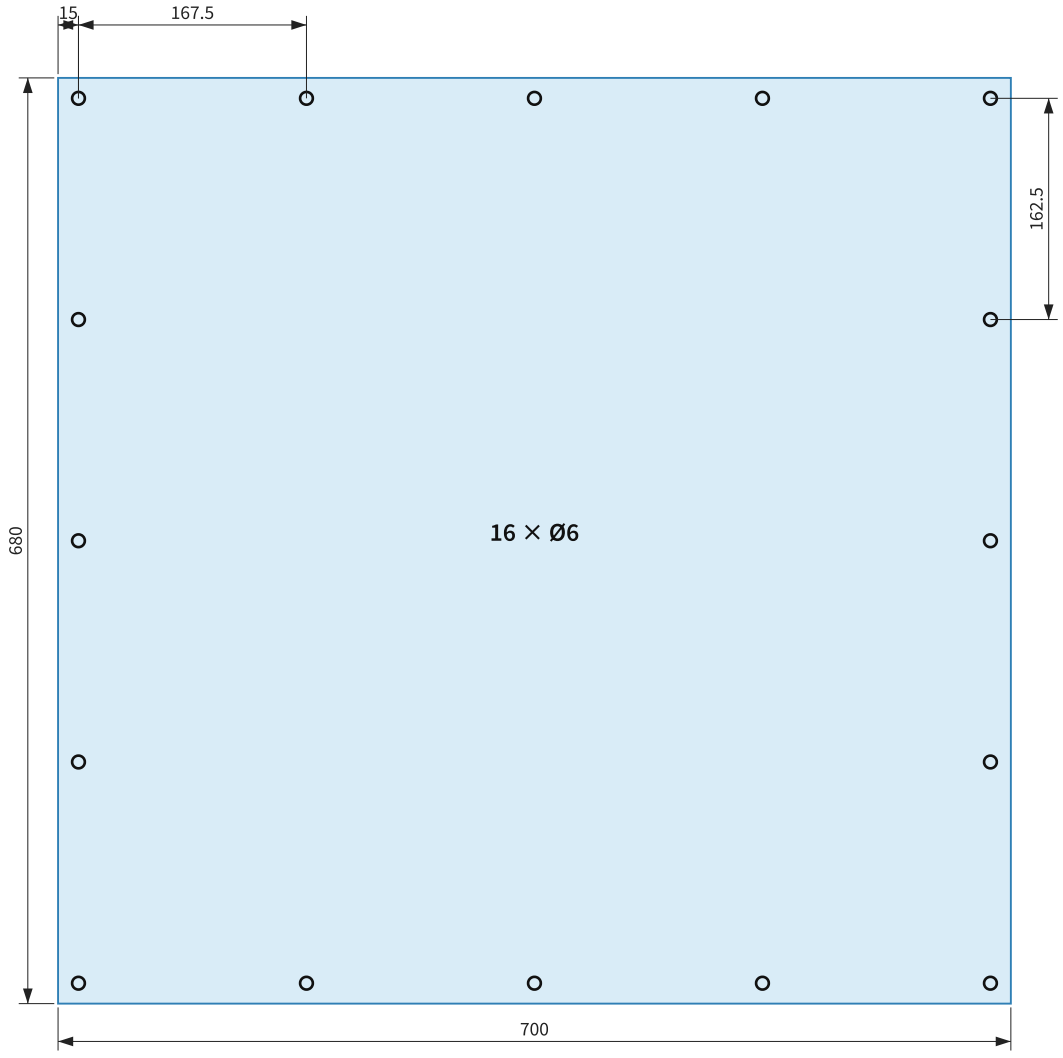
ID	Part	Size	Qty
A	Post	30×40×600	4
B	Front/back rail	30×40×640	4
C	Side rail	30×40×640	4
D	Door stile	30×40×683	2
E	Door rail	30×40×566	2
F	Base board	650×705×15	1

Strip total 10 018 mm (6 × 2 m); no beads.

Panel	Size	Qty	Pitch (long / short)
P1	700×680	2	167.5 / 162.5
P2	646×680	1	162.5 / 154
P3	646×703	1	168.25 / 154
P4	646×683	1	163.25 / 154

Each panel has 5 holes per edge (corners shared, 16 in all), Ø6, centres 15 from the edge. P2 Ø50 hole: 113 from the left and 95 from the bottom, seen from outside; optional Ø116 fan hole 118 from the right and 115 from the top. P3 Ø50 hole: 103 from the right, 113 from the top (rear) edge.

P1 hole layout 1:5.5 (others alike)



Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

Appendix: screw-on version

Sheet	Scale	Units	Date	Rev.
8 / 8	—	mm	2026-11-08	v2